Connected continuation of the smooth gCLM profiles and a locally unique analytic branch at the CLM anchor

Discovery note for arXiv:2305.05895

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Abstract

The source paper constructs, for every $a \leq 1$, a fixed point of a continuous map $\mathbf{R}_a$ on one compact convex set $\mathbb{D}$, and conjectures uniqueness, pointwise monotonicity in a, and a continuous choice of fixed points. We prove two partial results. First, on every compact parameter interval the fixed-point graph has a connected component projecting onto the whole interval; on $[0, 1]$ it joins the unique CLM and De Gregorio anchors, and every selection is continuous at both anchors. Second, the exact CLM spectrum recently computed in arXiv:2607.19762 makes the scale-fixed profile equation analytically invertible, giving a locally unique real-analytic branch of strongly regular normalized profiles near $a = 0$. We also explain why neither conclusion yet gives the source's global monotone graph.

1 Source conjecture and result

Let $\rho(x) = (1 + |x|)^{-1/2}$, let V be the source paper's Banach space with norm $\|f\|_\rho = \|\rho f\|_\infty$, and let $\mathbb{D} \subset V$ be its compact convex invariant set. The map $\mathbf{R}_a : \mathbb{D} \to \mathbb{D}$ is defined in Section 3 of the source for every $a \leq 1$. Put

$$\mathcal{F}_a = \mathrm{Fix}(\mathbf{R}_a), \qquad \mathcal{G}_I = \{(a, f) \in I \times \mathbb{D} : f = \mathbf{R}_a(f)\}$$

for a compact interval $I \subset (-\infty, 1]$. Conjecture 3.12 asks for pointwise parameter monotonicity of all these fixed points (hence uniqueness of every $\mathcal{F}_a$) and for a continuous selection $a \mapsto f_a$.

Theorem 1 (Unconditional continuation and anchor localization)**.** *For every compact interval $I = [A, B] \subset (-\infty, 1]$:*

1. *$\mathcal{G}_I$ is nonempty and compact, and it has a compact connected component C_I whose projection onto the parameter coordinate is all of I.*
2. *On $I = [0, 1]$, such a component meets $(0, f_0)$ and $(1, f_1)$, where f_0 and f_1 are the unique fixed points at the two endpoints.*
3. *In the Hausdorff metric of V,*

$$d_{\mathrm{H}}(\mathcal{F}_a, \{f_0\}) \longrightarrow 0 \quad (a \to 0), \qquad d_{\mathrm{H}}(\mathcal{F}_a, \{f_1\}) \longrightarrow 0 \quad (a \to 1^-).$$

Consequently every arbitrary choice $f_a \in \mathcal{F}_a$ is continuous at $a = 0$ and at $a = 1$.

Moreover the conjectured order is exactly true for the two explicit anchors $a = 0$ and $a = \frac{1}{2}$:

$$f_0(x) = \frac{1}{1 + x^2} \;\geq\; f_{1/2}(x) = \frac{4}{(2 + x^2)^2} \qquad (x \in \mathbb{R}).$$

The second result concerns the equivalent profile equation in the standard normalization with time exponent -1. Let $X = H^2_{\mathrm{odd}}(\mathbb{R})$ with the equivalent norm $\|\phi\|_X^2 = \|\phi\|_2^2 + \|\phi''\|_2^2$, and let

$$E = \{\Omega \in X : y\Omega' \in X\}$$

with its graph norm. For odd Ω, define $U'_\Omega = \mathcal{H}\Omega$ and $U_\Omega(0) = 0$, where $\mathcal{H}$ is the Hilbert transform, and set

$$\Phi(a, \Omega, c) = \Omega + (cy + aU_\Omega)\Omega' - \Omega\mathcal{H}\Omega.$$

The exact CLM point is

$$(a, \Omega, c) = (0, \Omega_0, 1), \qquad \Omega_0(y) = -\frac{y}{y^2 + \frac{1}{4}}.$$

Theorem 2 (Locally unique analytic CLM branch)**.** *There are $\varepsilon > 0$, a graph-norm neighborhood $\mathcal{U}$ of Ω_0 in E, and a neighborhood J of 1 such that, for every $|a| < \varepsilon$, the system*

$$\Phi(a, \Omega, c) = 0, \qquad \Omega'(0) = \Omega_0'(0) = -4$$

has exactly one solution $(\Omega_a, c_a) \in \mathcal{U} \times J$. The map $a \mapsto (\Omega_a, c_a)$ is real analytic.

This is local uniqueness in the strong profile graph norm. It does not assert that every Schauder fixed point in $\mathbb{D}$ lies in this neighborhood.

2 Idea of the proof

The source already supplies more than separate Schauder fixed points. Its compactness lemma makes $\mathbb{D}$ a compact convex set, and the proof of its uniform tail lemma records joint continuity of $(a, f) \mapsto \mathbf{R}_a(f)$, including the removable formula at $a = 0$. Browder's parametric strengthening of Schauder then forces one connected component of fixed points to cross any compact parameter interval. Uniqueness at an endpoint upgrades compactness to convergence of the entire fixed-point fiber there.

For the strong local statement, the profile residual is analytic on a graph domain. The exact spectral theorem at $a = 0$ says that the only zero mode of its linearization is spatial scaling and that this mode is semisimple. The unknown speed c supplies that missing spectral direction, while the trace $\Omega'(0) = -4$ fixes the scale. The augmented derivative is therefore an isomorphism, so the analytic implicit-function theorem applies.

3 Proof of Theorem 1

The source proves continuity of $f \mapsto \mathbf{R}_a(f)$ on $\mathbb{D}$. More importantly, in the proof of its uniform-tail Lemma 4.7 it observes that if $a_n \to a$ and $f_n \to f$ in V, then

$$\mathbf{R}_{a_n}(f_n) \longrightarrow \mathbf{R}_a(f) \quad \text{in } V.$$

At $a = 0$ this uses continuity of $(1-t)^{1/t}$ at zero. The estimates in that proof are uniform when a stays in a compact subinterval of $(-\infty, 1]$. Hence $(a, f) \mapsto \mathbf{R}_a(f)$ is jointly continuous on $I \times \mathbb{D}$.

The parameter space I is compact and connected and $\mathbb{D}$ is a nonempty compact convex subset of the locally convex space V. Browder's parametric fixed-point theorem therefore says that $\mathcal{G}_I$ has a connected component whose projection is I. Joint continuity also makes $\mathcal{G}_I$ closed in the compact set $I \times \mathbb{D}$, hence compact. This proves the first assertion. For $I = [0, 1]$, the source proves that the endpoint fibers are the singletons $\{f_0\}$ and $\{f_1\}$, so a component projecting onto the whole interval must meet both indicated points.

For anchor localization, suppose for example that the first Hausdorff limit fails. There are $a_n \to 0$ and $f_n \in \mathcal{F}_{a_n}$ with $\|f_n - f_0\|_\rho \geq \delta > 0$. Compactness of $\mathbb{D}$ gives a subsequence $f_n \to f_*$. Joint continuity and $f_n = \mathbf{R}_{a_n}(f_n)$ imply $f_* = \mathbf{R}_0(f_*)$, so endpoint uniqueness gives $f_* = f_0$, a contradiction. The proof at $a = 1$ is identical.

Finally, with $t = x^2$,

$$\frac{1}{1+x^2} - \frac{4}{(2+x^2)^2} = \frac{(t+2)^2 - 4(t+1)}{(1+t)(2+t)^2} = \frac{x^4}{(1+x^2)(2+x^2)^2} \geq 0.$$

4 Proof of Theorem 2

First we verify the nonlinear mapping. For $\Omega \in X$, put

$$Q_\Omega(y) = \frac{U_\Omega(y)}{y} = \int_0^1 (\mathcal{H}\Omega)(ty)\, dt,$$

with the evident value at $y = 0$. For $k = 0, 1, 2$, dilation and Minkowski give

$$\|\partial_y^k Q_\Omega\|_2 \leq \|\partial_y^k \mathcal{H}\Omega\|_2 \int_0^1 t^{k-1/2}\, dt.$$

Thus $\Omega \mapsto Q_\Omega$ is bounded on H^2. Since $H^2(\mathbb{R})$ is a Banach algebra and $U_\Omega \Omega' = Q_\Omega(y\Omega')$, the map

$$\Phi : \mathbb{R} \times E \times \mathbb{R} \longrightarrow X$$

is real analytic.

Let L_0 be the closed CLM linearization on X with domain E. Direct differentiation gives

$$D_\Omega \Phi(0, \Omega_0, 1) = -L_0, \qquad \partial_c \Phi(0, \Omega_0, 1) = s := y\Omega_0'.$$

The exact spectral results of arXiv:2607.19762 show that zero is an isolated semisimple eigenvalue of L_0, its eigenspace is one-dimensional and spanned by s, and the restriction of L_0 to the complementary Riesz subspace is boundedly invertible (from X into the graph domain). Let P be the Riesz projection at zero and write

$$X = \operatorname{span}\{s\} \oplus X_1, \qquad E = \operatorname{span}\{s\} \oplus E_1.$$

The derivative of the augmented map $(\Phi, \Omega'(0) + 4)$ is

$$\mathcal{A}(\phi, d) = (-L_0\phi + ds, \phi'(0)).$$

It is an isomorphism. Indeed, given $(g, r) \in X \times \mathbb{R}$, decompose $g = \beta s + g_1$ with $g_1 \in X_1$, take

$$d = \beta, \qquad \phi_1 = -\big(L_0|_{E_1}\big)^{-1} g_1, \qquad \phi = \alpha s + \phi_1,$$

and choose α so that $\phi'(0) = r$. This is possible and unique since

$$s'(0) = \Omega_0'(0) = -4 \neq 0.$$

All operations are bounded; uniqueness follows from the same decomposition. The analytic implicit-function theorem now proves the claim.

5 Why this is not yet the full conjecture

The kernel of the source's linear transform is

$$K(x,y) = \frac{y}{x}\log\left|\frac{x+y}{x-y}\right| - 2.$$

For $t = y/x$, it tends to -2 as $t \to 0$ but equals $2/(3t^2) + O(t^{-4}) > 0$ as $t \to \infty$. Thus $\mathbf{T}$ is not positivity preserving, blocking a direct order iteration for the pointwise conjecture. The exact linearized inverse at $a = 0$ is strongly nonnormal and has no known positivity property either.

Nor does a connected fixed-point continuum automatically contain a continuous section. On $[0,1] \times [0,1]$, set

$$g(t) = \frac{1+\sin(1/t)}{2} \quad (t > 0), \qquad F(t,x) = (1-t)x + tg(t), \qquad F(0,x) = x.$$

This is jointly continuous. Its fixed set is

$$\{(t, g(t)) : t > 0\} \cup (\{0\} \times [0,1]),$$

the topologist's sine curve: it is connected and projects onto the whole parameter interval, but it has no continuous section. A proof of the full source conjecture therefore needs additional analytic sign or strong compactness structure.

References

1. D. Huang, X. Qin, X. Wang, and D. Wei, *Self-similar finite-time blowups with smooth profiles of the generalized Constantin–Lax–Majda model*, Arch. Ration. Mech. Anal. 248 (2024), article 22; arXiv:2305.05895. Conjecture 3.12 and Sections 3–5.

2. E. Solan and O. N. Solan, *Browder's Theorem: from One-Dimensional Parameter Space to General Parameter Space*, arXiv:2210.16369. Parametric fixed-point theorem for compact connected parameter spaces and compact convex subsets of locally convex spaces.

3. J. Xu, *The spectral picture of self-similar collapse in the Constantin–Lax–Majda equation*, arXiv:2607.19762 (2026). Exact origin-H^2 spectrum, simple symmetry modes, and resolvent at $a = 0$.